

32. Compare Match Timer W (CMTW)

This MCU includes two units (unit 0 and unit 1) with one channel of 32-bit compare match timer W (CMTW). CMTW has a 32-bit counter and can generate interrupts each time a set period elapses.

In this section, “PCLK” is used to refer to PCLKB.

32.1 Overview

Table 32.1 shows the specifications of the CMTW.

Figure 32.1 shows a block diagram of the CMTW0 and Figure 32.2 shows a block diagram of the CMTW1.

Table 32.1 CMTW Specifications

Item	Function
Number of channels	Two channels (unit 0, unit 1)
Timer counter	16-bit/32-bit selectable up-counter The counter returns to 0000 0000h after a compare match.
Prescaler	Four dividing clocks are output. Selectable from any of PCLK/8, PCLK/32, PCLK/128, and PCLK/512
Input capture	Up to two input capture input signals available.
Output compare	Up to two output compare output signals available.
Compare match	One compare match available (no output compare output pin used).
Interrupts	Compare match interrupt Input capture 0 and 1 interrupts Output compare 0 and 1 interrupts
Event link function (output) (Unit 0)	Compare match
Event link function (input) (Unit 0)	One of the following operations is enabled after an event is accepted: - Count start operation - Event count operation - Count restart operation
Low power consumption function	Each unit can be placed in the module stop state.

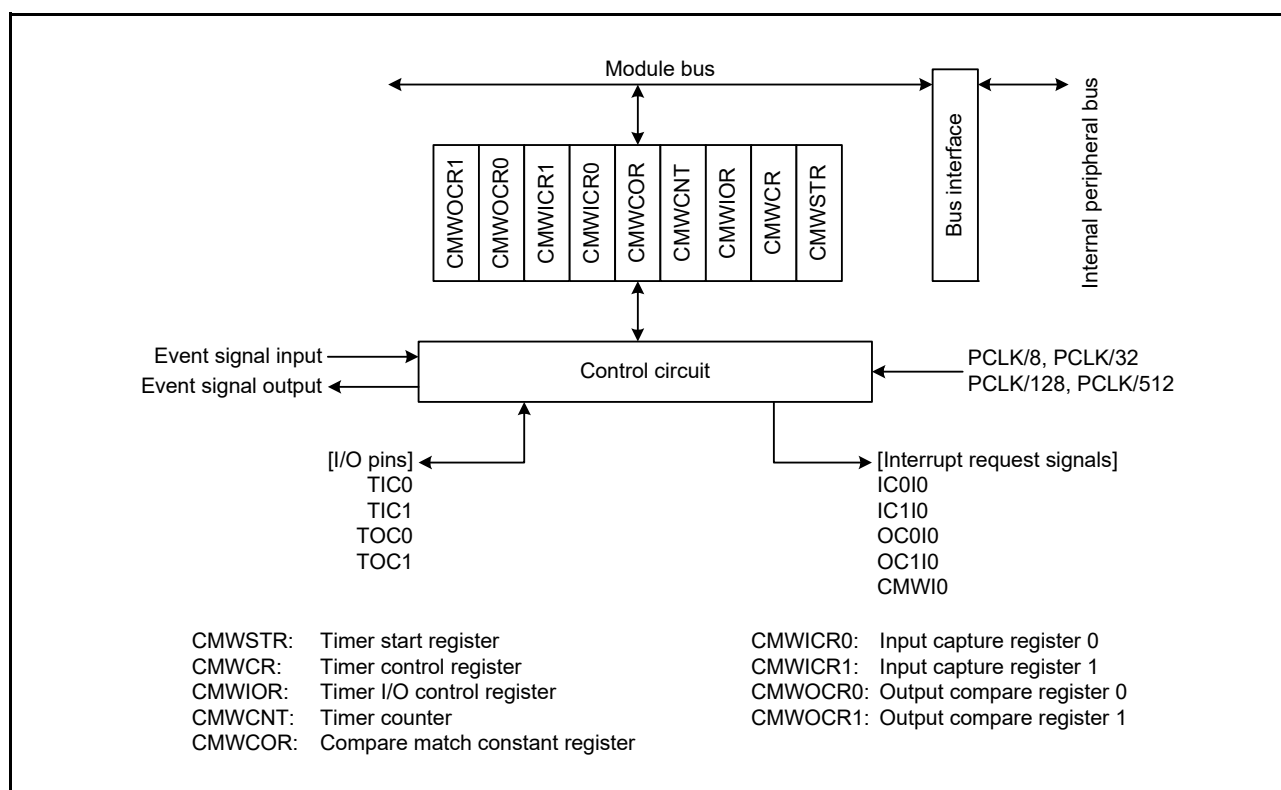


Figure 32.1 CMTW0 Block Diagram

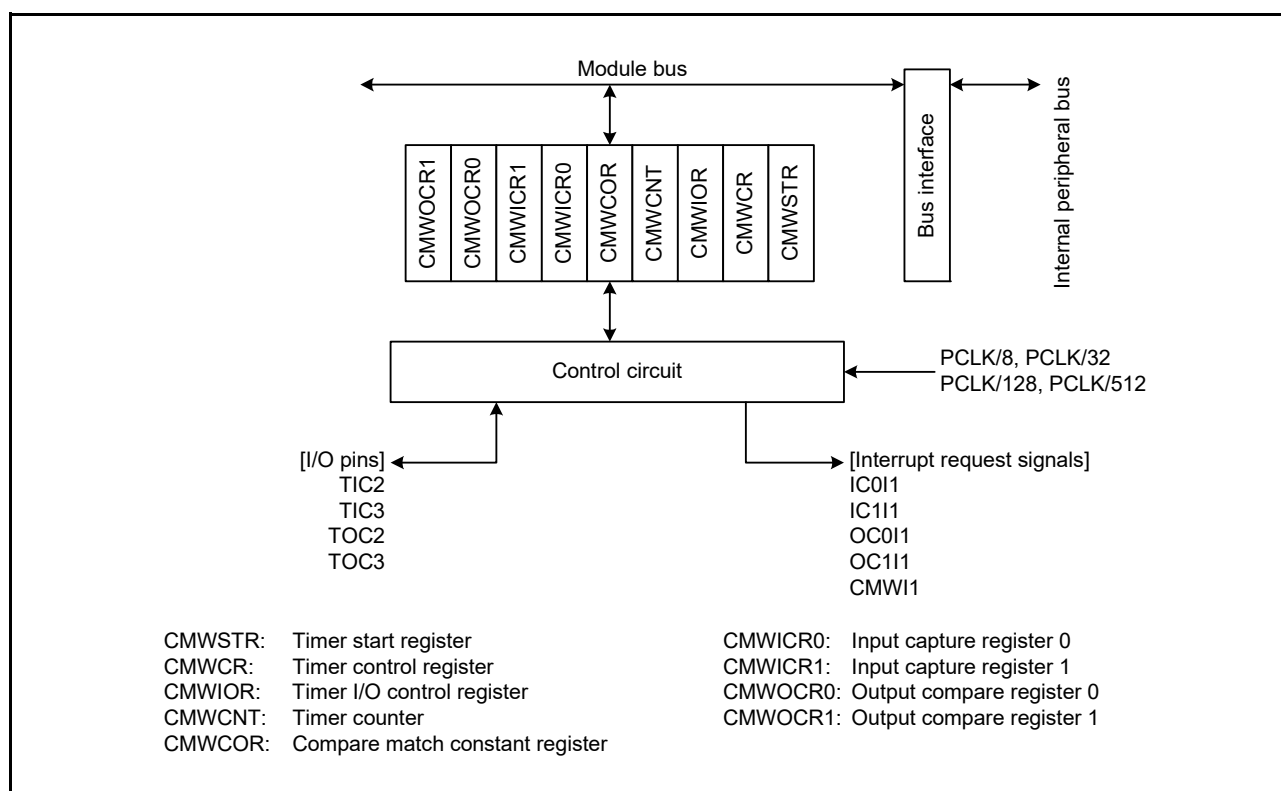


Figure 32.2 CMTW1 Block Diagram

Table 32.2 shows the CMTW pin configuration.

Table 32.2 CMTW Pin Configuration

Unit	Pin Name	I/O	Description
CMTW0	TIC0	Input	Input capture input for the CMTW0.CMWICR0 register
	TIC1	Input	Input capture input for the CMTW0.CMWICR1 register
	TOC0	Output	Output compare output for the CMTW0.CMWOCR0 register
	TOC1	Output	Output compare output for the CMTW0.CMWOCR1 register
CMTW1	TIC2	Input	Input capture input for the CMTW1.CMWICR0 register
	TIC3	Input	Input capture input for the CMTW1.CMWICR1 register
	TOC2	Output	Output compare output for the CMTW1.CMWOCR0 register
	TOC3	Output	Output compare output for the CMTW1.CMWOCR1 register

32.2 Register Descriptions

32.2.1 Timer Start Register (CMWSTR)

Address(es): CMTW0.CMWSTR 0009 4200h, CMTW1.CMWSTR 0009 4280h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	STR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	STR	Counter Start	0: CMWCNT counter count is stopped. (The value immediately before count operation stops is retained and the count operation is stopped.) 1: CMWCNT counter count is started.	R/W
b15 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

STR Bit (Counter Start)

Specifies whether the timer counter operates or is stopped. The relevant prescaler operates or is stopped according to the settings of the STR bit.

32.2.2 Timer Control Register (CMWCR)

Address(es): CMTW0.CMWCR 0009 4204h, CMTW1.CMWCR 0009 4284h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
CCLR[2:0]	—	—	—	CMS	—	OC1IE	OC0IE	IC1IE	IC0IE	CMWIE	—	CKS[1:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	CKS[1:0]	Clock Select	b1 b0 0 0: PCLK/8 0 1: PCLK/32 1 0: PCLK/128 1 1: PCLK/512	R/W
b2	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b3	CMWIE	Compare Match Interrupt Request Enable	0: Interrupt request (CMWI) disabled 1: Interrupt request (CMWI) enabled	R/W
b4	IC0IE	Input Capture 0 Interrupt Request Enable	0: Interrupt request (IC0I) disabled 1: Interrupt request (IC0I) enabled	R/W
b5	IC1IE	Input Capture 1 Interrupt Request Enable	0: Interrupt request (IC1I) disabled 1: Interrupt request (IC1I) enabled	R/W
b6	OC0IE	Output Compare 0 Interrupt Request Enable	0: Interrupt request (OC0I) disabled 1: Interrupt request (OC0I) enabled	R/W
b7	OC1IE	Output Compare 1 Interrupt Request Enable	0: Interrupt request (OC1I) disabled 1: Interrupt request (OC1I) enabled	R/W
b8	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b9	CMS	Timer Counter Size	0: 32 bits 1: 16 bits	R/W
b12 to b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b15 to b13	CCLR[2:0]	Counter Clear	b15 b13 0 0 0: CMWCNT counter cleared by CMWCOR register compare match 0 0 1: Clearing of CMWCNT counter disabled 0 1 0: Clearing of CMWCNT counter disabled 0 1 1: Clearing of CMWCNT counter disabled 1 0 0: CMWCNT counter cleared by CMWICR0 register input capture 1 0 1: CMWCNT counter cleared by CMWICR1 register input capture 1 1 0: CMWCNT counter cleared by CMWOCR0 register compare match 1 1 1: CMWCNT counter cleared by CMWOCR1 register compare match	R/W

The CMWCR register should be set while the timer counter (CMWCNT) operation is stopped.

CKS[1:0] Bits (Clock Select)

Select the clock to be input to the CMWCNT counter among four internal clocks obtained by dividing the peripheral module clock (PCLK). When the CMWSTR.STR bit is set to 1, the CMWCNT counter starts counting based on the clock selected with the CMWCR.CKS[1:0] bits.

CMWIE Bit (Compare Match Interrupt Request Enable)

Enables or disables compare match interrupt request (CMWI) generation when the CMWCNT counter and the CMWCOR register values match.

IC0IE Bit (Input Capture 0 Interrupt Request Enable)

Enables or disables input capture 0 interrupt request (IC0I) generation when input capture is generated in the CMWICR0 register.

IC1IE Bit (Input Capture 1 Interrupt Request Enable)

Enables or disables input capture 1 interrupt request (IC1I) generation when input capture is generated in the CMWICR1 register.

OC0IE Bit (Output Compare 0 Interrupt Request Enable)

Enables or disables output compare 0 interrupt request (OC0I) generation when the CMWCNT counter and the CMWOCR0 register values match.

OC1IE Bit (Output Compare 1 Interrupt Request Enable)

Enables or disables output compare 1 interrupt request (OC1I) generation when the CMWCNT counter and CMWOCR1 register values match.

CMS Bit (Timer Counter Size)

Selects either 16 or 32 bits as the size of the timer counter (CMWCNT). The size selected with the CMS bit is valid in the CMWCOR, CMWICR0, CMWICR1, CMWOCR0, and CMWOCR1 registers.

CCLR[2:0] Bits (Counter Clear)

Select the CMWCNT counter clearing source.

32.2.3 Timer I/O Control Register (CMWIOR)

Address(es): CMTW0.CMWIOR 0009 4208h, CMTW1.CMWIOR 0009 4288h

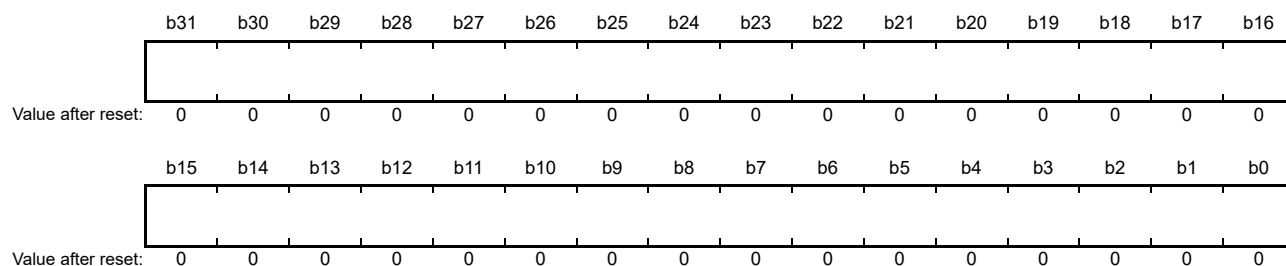
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	CMWE	—	OC1E	OC0E	OC1[1:0]	OC0[1:0]	—	—	IC1E	IC0E	IC1[1:0]	IC0[1:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	IC0[1:0]	Input Capture 0 Control	b1 b0 0 0: Input capture at the rising edge 0 1: Input capture at the falling edge 1 0: Input capture at both edges 1 1: Setting prohibited	R/W
b3, b2	IC1[1:0]	Input Capture 1 Control	b3 b2 0 0: Input capture at the rising edge 0 1: Input capture at the falling edge 1 0: Input capture at both edges 1 1: Setting prohibited	R/W
b4	IC0E	Input Capture 0 Enable	0: Input capture 0 operation disabled 1: Input capture 0 operation enabled	R/W
b5	IC1E	Input Capture 1 Enable	0: Input capture 1 operation disabled 1: Input capture 1 operation enabled	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b9, b8	OC0[1:0]	Output Compare 0 Control	b9 b8 0 0: Retains the output value.*1 0 1: Initially outputs low and toggles the output value upon compare match. 1 0: Initially outputs high and toggles the output value upon compare match. 1 1: Setting prohibited	R/W
b11, b10	OC1[1:0]	Output Compare 1 Control	b11 b10 0 0: Retains the output value.*1 0 1: Initially outputs low and toggles the output value upon compare match. 1 0: Initially outputs high and toggles the output value upon compare match. 1 1: Setting prohibited	R/W
b12	OC0E	Output Compare 0 Enable	0: Output compare 0 operation disabled 1: Output compare 0 operation enabled	R/W
b13	OC1E	Output Compare 1 Enable	0: Output compare 1 operation disabled 1: Output compare 1 operation enabled	R/W
b14	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b15	CMWE	Compare Match Enable	0: Compare match operation disabled 1: Compare match operation enabled	R/W

Note 1. After reset, low is output until the CMWIOR register is set.

32.2.4 Timer Counter (CMWCNT)

Address(es): CMTW0.CMWCNT 0009 4210h, CMTW1.CMWCNT 0009 4290h



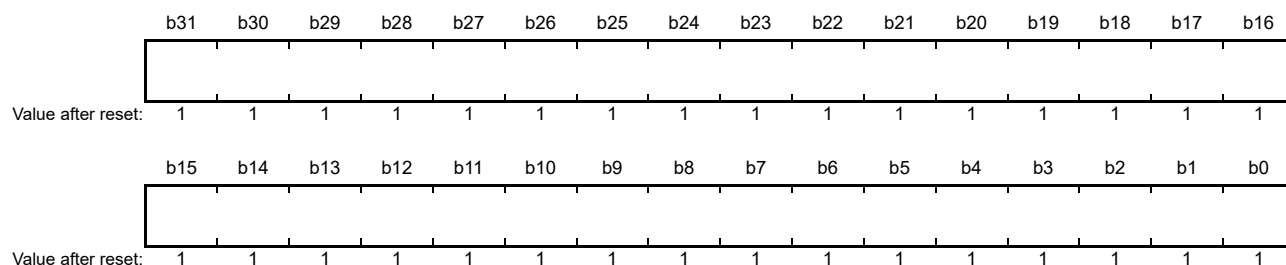
The CMWCNT counter is a readable/writable up-counter.

Before starting count operation, the timer control register (CMWCR) should be set. When the 16-bit counter size is selected with the CMS bit in the timer control register (CMWCR), bits 15 to 0 in this register are valid. When writing to the CMWCNT counter, write data in 32-bit units with the upper bits set to 0000h. The CMWCNT counter can only be accessed in longword units.

When the CMWSTR.STR bit is set to 1, the CMWCNT counter starts counting. When the CMWSTR.STR bit is set to 0, the CMWCNT counter retains the value immediately before a stop of counting and stops counting.

32.2.5 Compare Match Constant Register (CMWCOR)

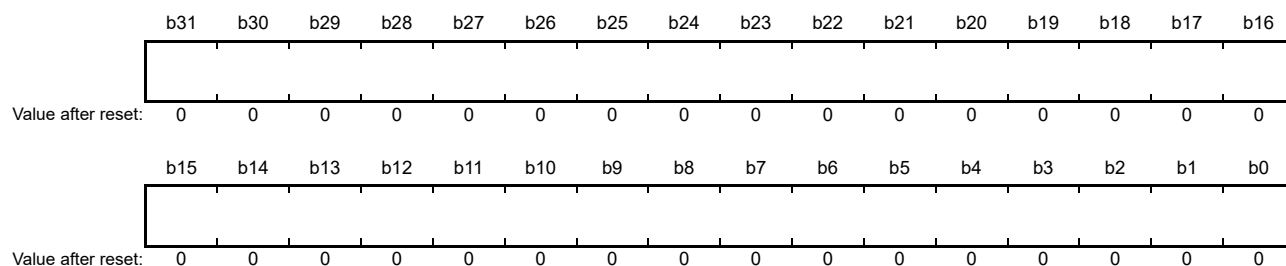
Address(es): CMTW0.CMWCOR 0009 4214h, CMTW1.CMWCOR 0009 4294h



The CMWCOR register is a readable/writable register that specifies the time up to a compare match between the timer counter (CMWCNT) value and CMWCOR value. When the 16-bit counter size is selected with the CMS bit in the timer control register (CMWCR), bits 15 to 0 in this register are valid. When writing to the CMWCOR register, write data in 32-bit units with the upper bits set to 0000h. The CMWCOR register can only be accessed in longword units. To detect an overflow, set the CMWCOR register value to FFFF FFFFh (32-bit count operation) or 0000 FFFFh (16-bit count operation). When the CMWCNT counter is set to 0, a compare match interrupt request (CMWI) can be used as an overflow detection signal.

32.2.6 Input Capture Register n (CMWICRn) (n = 0, 1)

Address(es): CMTW0.CMWICR0 0009 4218h, CMTW0.CMWICR1 0009 421Ch, CMTW1.CMWICR0 0009 4298h, CMTW1.CMWICR1 0009 429Ch

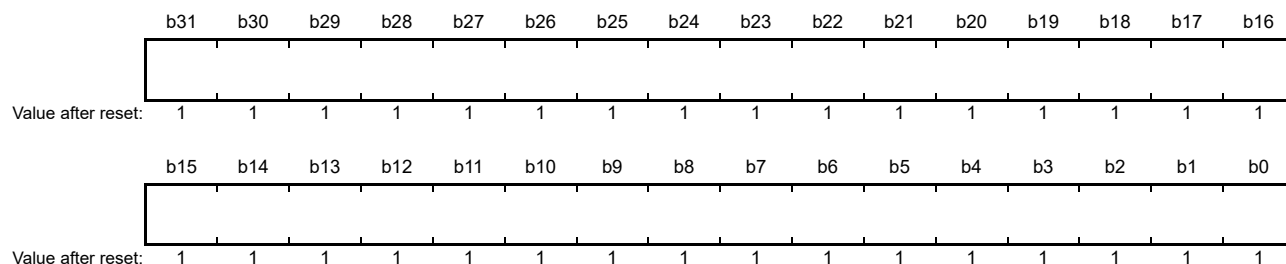


The CMWICRn register is a read-only register in which the CMWCNT value is stored when an input capture is generated.

When the 16-bit counter size is selected with the CMWCR.CMS bit, bits 15 to 0 in these registers are valid. Writing to these registers is invalid. The CMWICRn register can only be accessed in longword units.

32.2.7 Output Compare Register n (CMWOCRn) (n = 0, 1)

Address(es): CMTW0.CMWOCR0 0009 4220h, CMTW0.CMWOCR1 0009 4224h, CMTW1.CMWOCR0 0009 42A0h, CMTW1.CMWOCR1 0009 42A4h



The CMWOCRn register is a readable/writable register that set the value to be compared when an output compare is generated.

When the 16-bit counter size is selected with the CMWCR.CMS bit, bits 15 to 0 of these registers become valid. When writing to these registers, write data in 32-bit units with the upper bits set to 0000h.

The CMWOCRn register can only be accessed in longword units. The initial value of CMWOCR0 and CMWOCR1 registers is FFFF FFFFh.

32.3 Operation

When the CMWCR register is set and then the STR bit in CMWSTR is set to 1, the CMTW starts count operation. Setting the CMWSTR.STR bit to 0 enables the CMWCNT counter to retain the value immediately before a stop of counting and stop counting. Setting the CMWIOR register enables using the compare match function, input capture input function, and output compare output function.

32.3.1 Period Count Operation

When the counter clock is selected by using the CMWCR.CKS[1:0] bits and the CMWSTR.STR bit is set to 1, the CMWCNT counter starts counting cycles of the selected clock. When clearing of the counter is selected by the CMWCR.CCLR[2:0] bits and the counter clearing source is generated, the CMWCNT counter becomes 0000 0000h and continues counting. When clearing of the counter is not selected, an overflow is generated when FFFF FFFFh changes to 0000 0000h during 32-bit count operation and 0000 FFFFh changes to 0000 0000h during 16-bit count operation, and the CMWCNT counter continues counting.

32.3.2 Compare Match Function

When the values of the CMWCNT counter and CMWCOR register match, the CMWCNT counter becomes 0000 0000h. At this time, a compare match interrupt request (CMWI) is generated. The CMWCNT counter restarts counting from 0000 0000h.

To enable overflow detection, the CMWCOR register value should be set to FFFF FFFFh (when the counter size is 32 bits) or 0000 FFFFh (when the counter size is 16 bits). When the values of the CMWCNT counter and CMWCOR register match, the CMWCNT counter becomes 0000 0000h. In this case, the compare match interrupt request (CMWI) is generated. The CMWCNT counter then restarts counting from 0000 0000h.

Figure 32.3 shows an example of procedure for setting compare match operation.

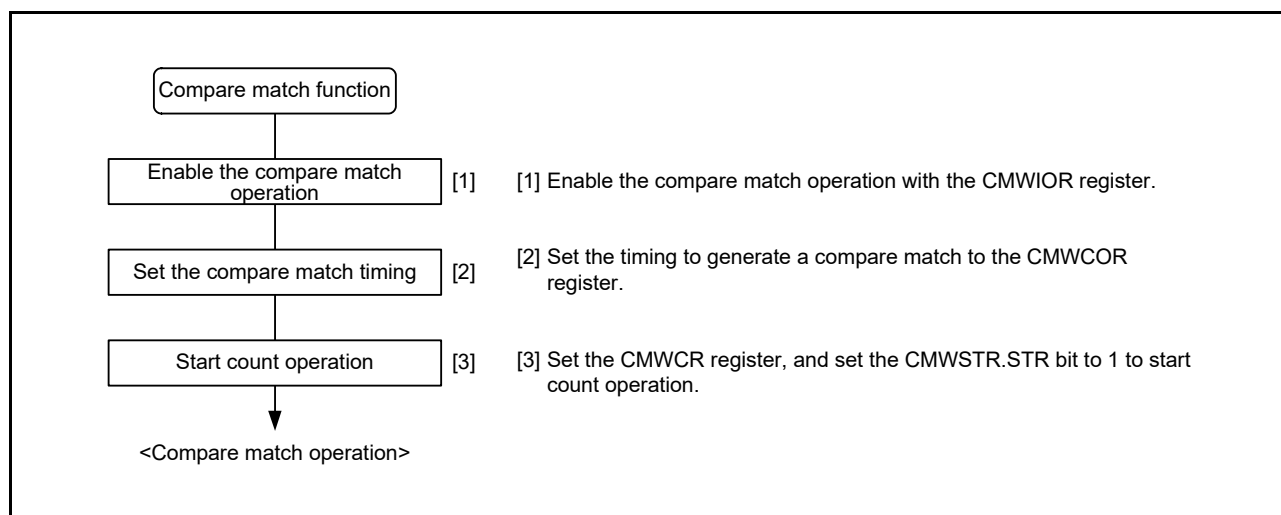


Figure 32.3 Procedure for Setting Compare Match Operation

Figure 32.4 shows an example when compare match with CMWCOR is set as a counter clearing source.

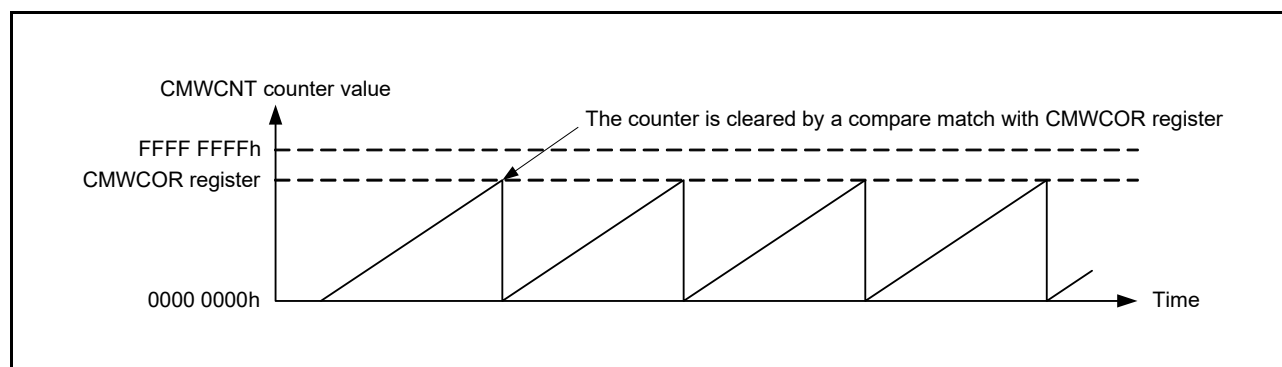


Figure 32.4 Example of Compare Match Operation

Figure 32.5 shows an example when CMWCOR is set to FFFF FFFFh and an overflow is detected.

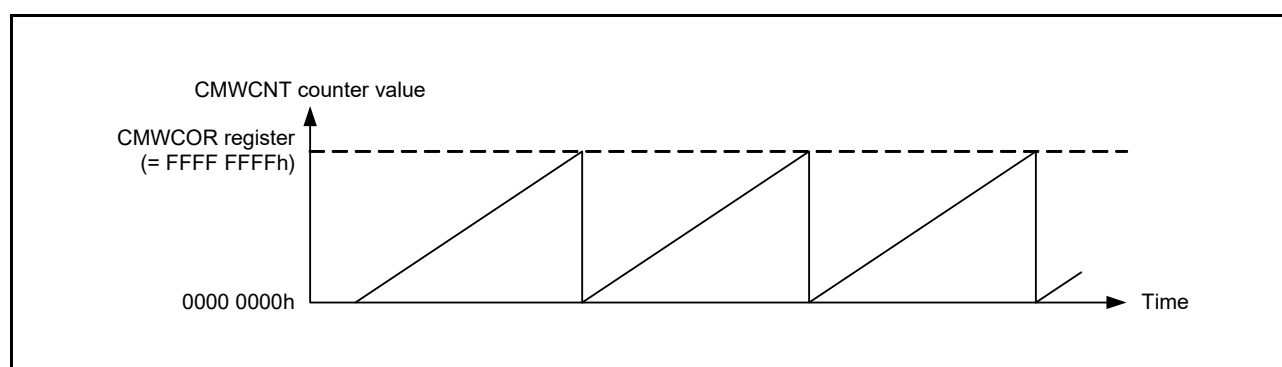


Figure 32.5 Example of Compare Match Operation (Overflow Detected)

32.3.3 Output Compare Function

The output compare function can be used for toggle waveform output. When the CMWCNT counter value matches either of the values of the CMWOCR0 or CMWOCR1 register, the output compare interrupt request (OC0I or OC1I) is generated. Figure 32.6 shows an example of procedure for setting output compare operation.

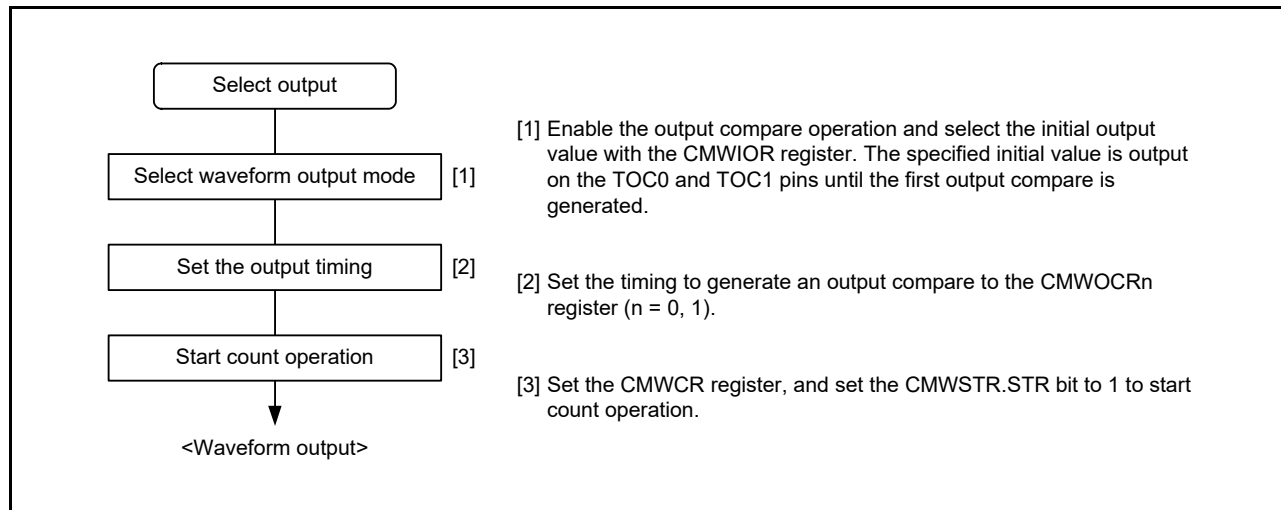


Figure 32.6 Procedure for Setting Output Compare Operation

Figure 32.7 shows an example of toggle waveform output from the TOC0 and TOC1 pins when the counter is set to be cleared by compare match with the CMWOCR1 register.

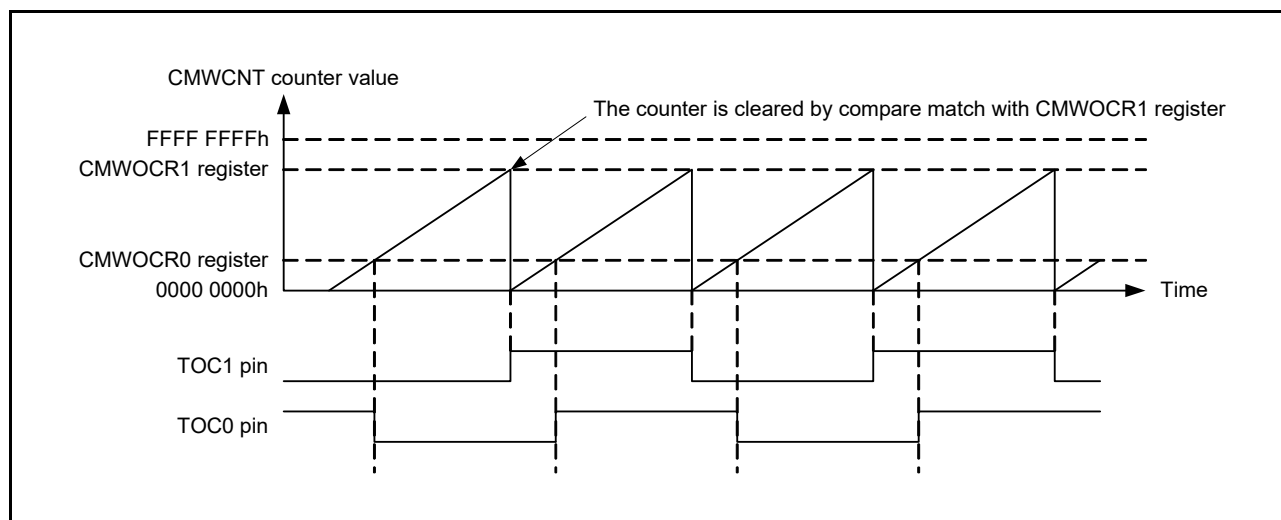


Figure 32.7 Example of Output Compare Operation (Unit 0)

32.3.4 Input Capture Function

Through detecting the edge on the TIC0 and TIC1 pin input, the CMWCNT counter value can be transferred to the CMWICR0 and CMWICR1 registers, respectively. The edges to be detected can be selected from among the rising edge alone, falling edge alone, and both the rising and falling edges. When the CMWCNT counter value is transferred to the CMWICR0 or CMWICR1 register using the input capture operation, an input capture interrupt request (IC0I or IC1I) is generated. Figure 32.8 shows an example of procedure for setting input capture operation.

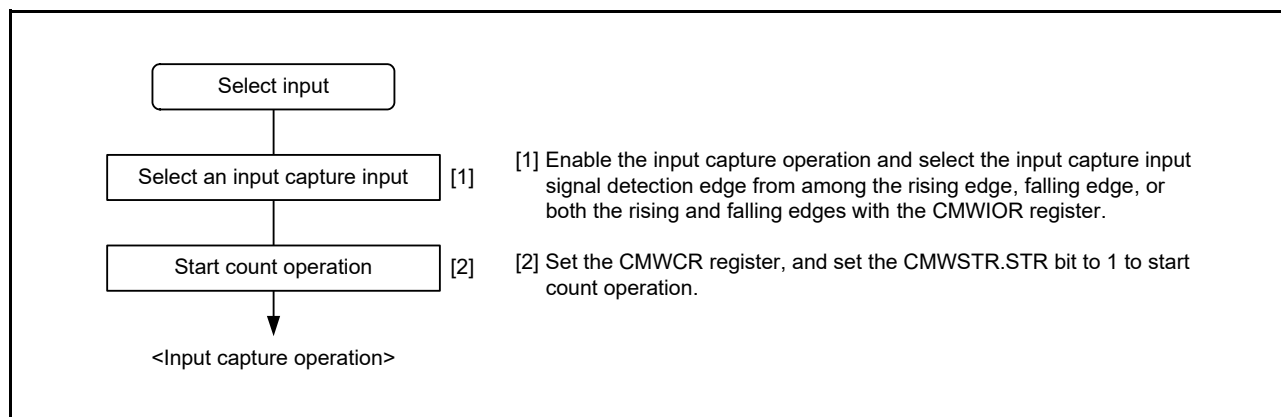


Figure 32.8 Procedure for Setting Input Capture Operation

Figure 32.9 shows an example in which both edges are selected for the TIC0 pin input capture detection edge and the falling edge for the TIC1 pin, and the CMWCNT counter is cleared by a CMWICR1 register input capture.

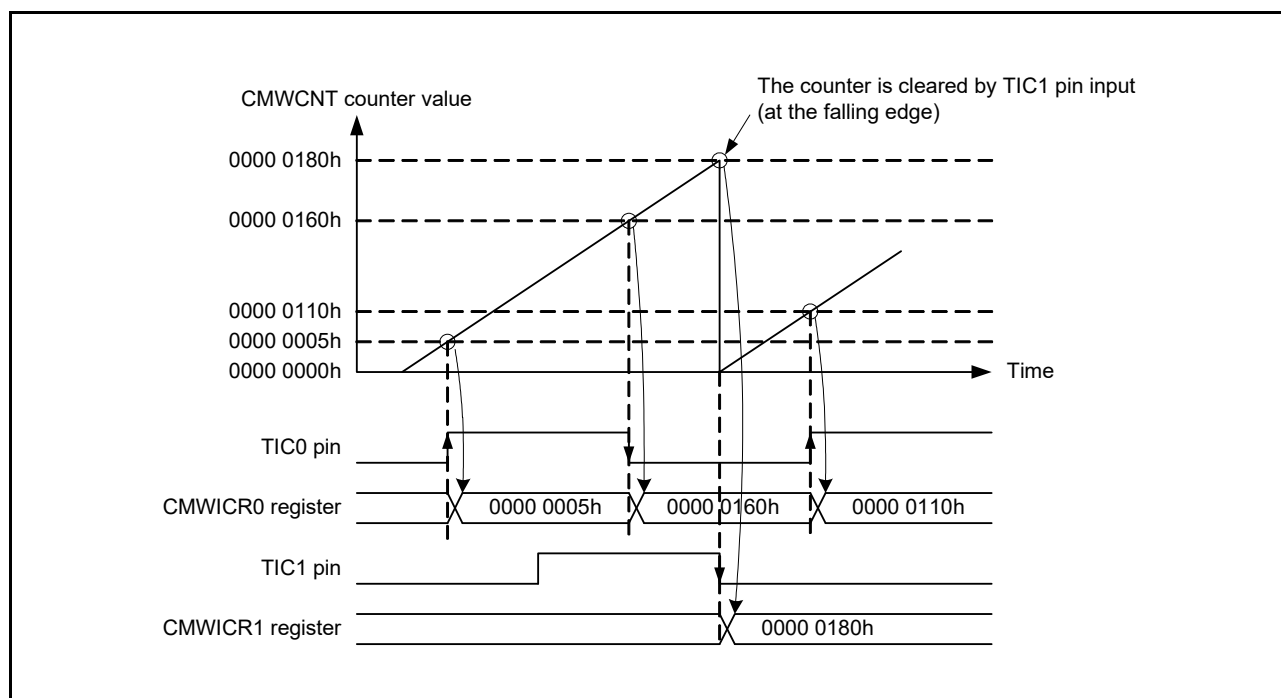


Figure 32.9 Example of Input Capture Operation (Unit 0)

32.3.5 Counter Size

With the CMTW, either 16 or 32 bits can be selected as the counter size by using the CMWCR.CMS bit.

When the counter is used as a 16-bit counter, set the value of the CMWCOR register in 32-bit units with the upper 16 bits set to 0000h. 0000 FFFFh should be set to detect an overflow. Similarly, set the values of the CMWOCR0 and CMWOCR1 registers in 32-bit units with the upper 16 bits set to 0000h. Read the CMWOCR0 and CMWOCR1 registers in 32-bit units. The upper 16 bits can be read as 0000h.

32.3.6 Count Timing of CMWCNT Counter

By setting the CMWCR.CKS[1:0] bits, one of four clocks (PCLK/8, PCLK/32, PCLK/128, and PCLK/512) obtained by dividing the peripheral module clock (PCLK) can be selected as the counter clock to be input to the CMWCNT counter. Figure 32.10 shows the timing.

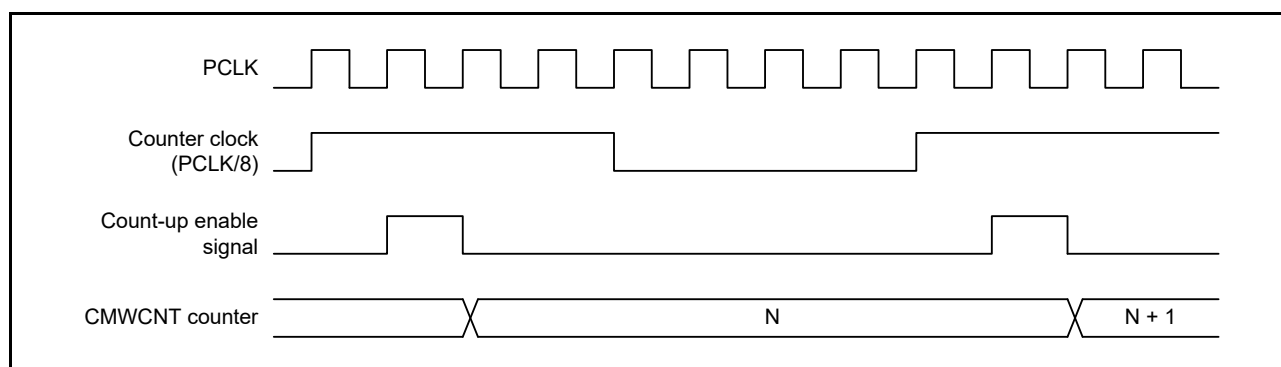


Figure 32.10 Count Timing (PCLK/8)

32.3.7 Output Compare Output Timing

A compare match signal is generated in the last state in which the CMWOCRn register ($n = 0, 1$) and CMWCNT counter values match (the CMWCNT counter value is updated immediately after the state). The compare match signal is generated if the CMWCNT count-up enable signal is input after a match between the CMWOCRn register and CMWCNT counter values. When a compare match signal is generated, output of the output compare pin (TOC pin) is toggled.

Figure 32.11 shows output compare output timing.

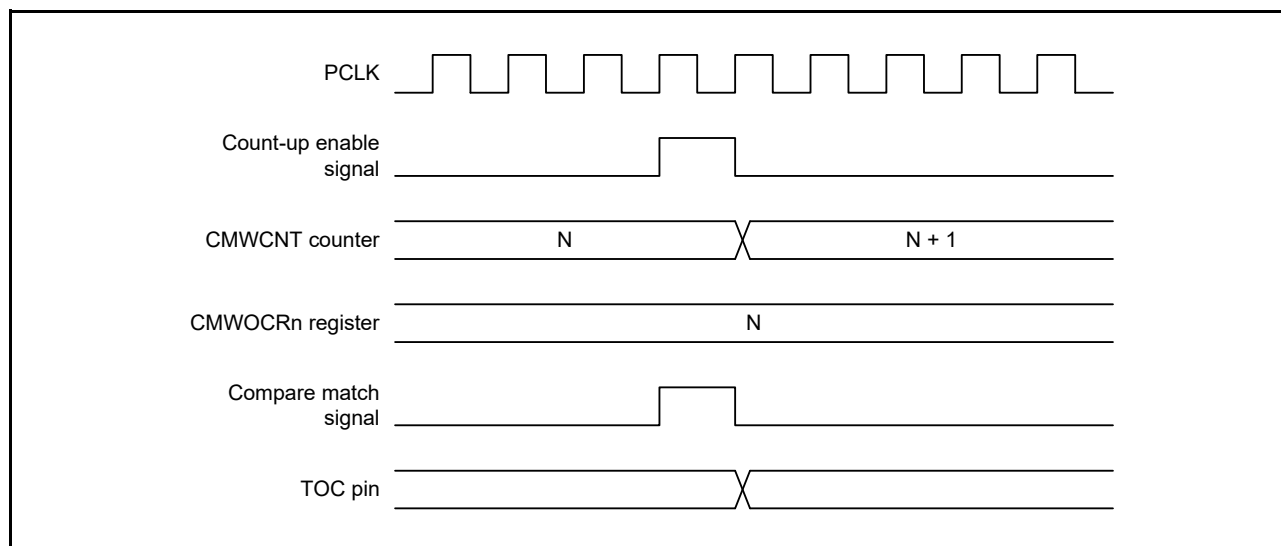


Figure 32.11 Output Compare Output Timing

32.3.8 Input Capture Timing

Figure 32.12 shows the timing of input capture operation at both edges.

When the edge of the TIC0 and TIC1 pins is detected, the CMWCNT counter value is transferred to the CMWICR0 and CMWICR1 registers, respectively.

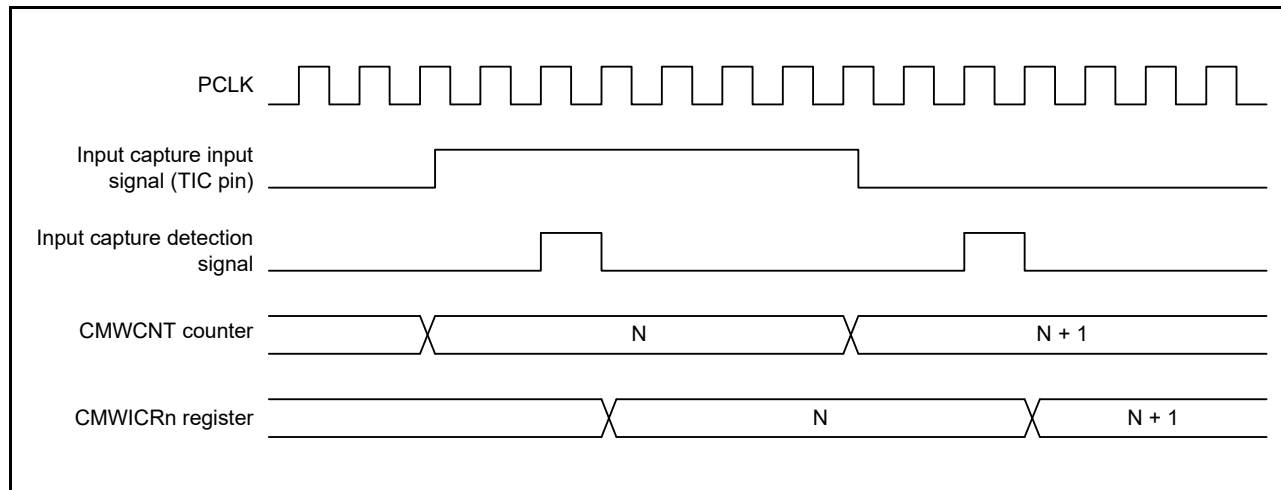


Figure 32.12 Input Capture Timing (Unit 0, Both-Edge Detection) (n = 0, 1)

32.4 Interrupts

32.4.1 CMTW Interrupt Sources and DTC/DMAC Transfer Requests

The CMTW has five interrupt sources: two input capture interrupt requests (IC0I and IC1I), two output compare interrupt requests (OC0I and OC1I), and a compare match interrupt request (CMWI).

Table 32.3 shows the interrupt sources. The interrupt sources can be enabled or disabled using the IC0IE, IC1IE, OC0IE, OC1IE, and CMWIE bits in CMWCR and are separately generated to the interrupt controller.

Each interrupt request can activate the DMAC or DTC. When the DMAC is used for data transfer, an interrupt request is not generated to the CPU. For generating an interrupt request to the CPU during data transfer using the DTC, refer to section 20, Data Transfer Controller (DTCb).

Table 32.3 CMTW Interrupt Sources

Unit	Name	Interrupt Request	Interrupt Request Enable Bit	DMAC/DTC Activation
CMTW0	CMWI0	Compare match of CMTW0.CMWCR0 register	CMTW0.CMWCR.CMWIE	Possible
	IC0I0	Input capture of CMTW0.CMWICR0 register	CMTW0.CMWCR.IC0IE	Possible
	IC1I0	Input capture of CMTW0.CMWICR1 register	CMTW0.CMWCR.IC1IE	Possible
	OC0I0	Output compare of CMTW0.CMWOCR0 register	CMTW0.CMWCR.OC0IE	Possible
	OC1I0	Output compare of CMTW0.CMWOCR1 register	CMTW0.CMWCR.OC1IE	Possible
CMTW1	CMWI1	Compare match of CMTW1.CMWCR0 register	CMTW1.CMWCR.CMWIE	Possible
	IC0I1	Input capture of CMTW1.CMWICR0 register	CMTW1.CMWCR.IC0IE	Possible
	IC1I1	Input capture of CMTW1.CMWICR1 register	CMTW1.CMWCR.IC1IE	Possible
	OC0I1	Output compare of CMTW1.CMWOCR0 register	CMTW1.CMWCR.OC0IE	Possible
	OC1I1	Output compare of CMTW1.CMWOCR1 register	CMTW1.CMWCR.OC1IE	Possible

32.4.2 Timing of Compare Match Interrupt Generation

When the values of the CMWCNT counter and CMWCOR register match, a compare match interrupt request (CMWI) is generated. The compare match signal is generated at the end of the cycle where the values matched (i.e. when the CMWCNT counter is updated from the matching counter value). The compare match signal, therefore, is not generated until a count-up enable signal is generated after the values of the CMWCNT counter and CMWCOR register have matched. Figure 32.13 shows the timing of compare match interrupt generation.

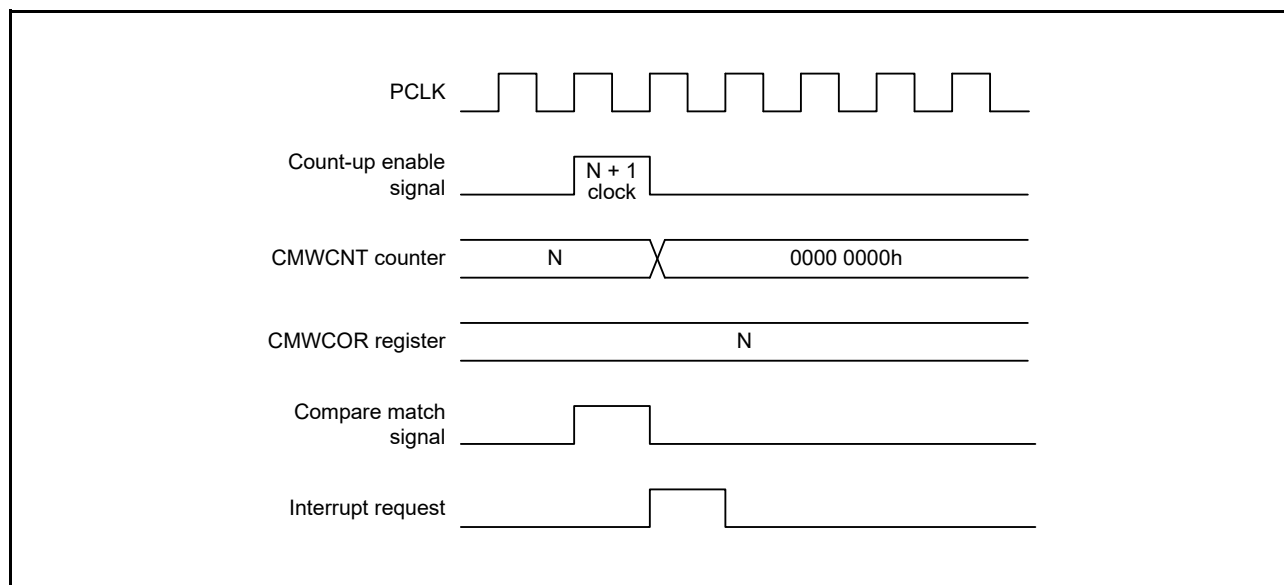


Figure 32.13 Timing of Compare Match Interrupt Generation

(a) Timing of Output Compare Interrupt Generation

Figure 32.14 shows the timing of output compare interrupt generation.

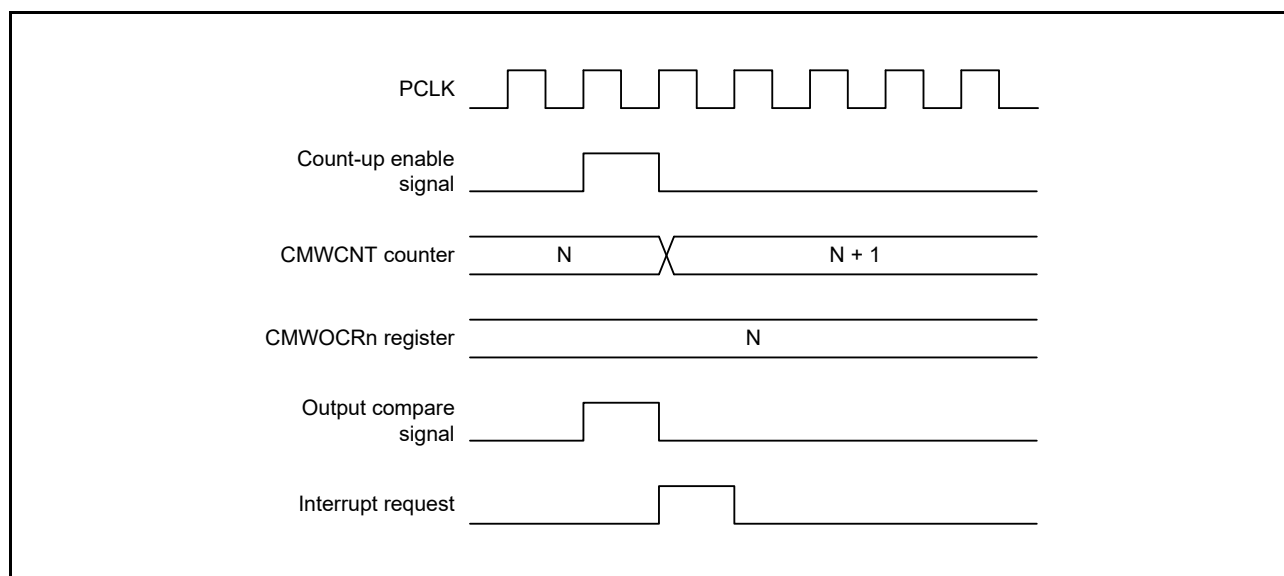


Figure 32.14 Timing of Output Compare Interrupt Generation ($n = 0, 1$)

(b) Timing of Input Capture Interrupt Generation

Figure 32.15 shows the timing of input capture interrupt generation.

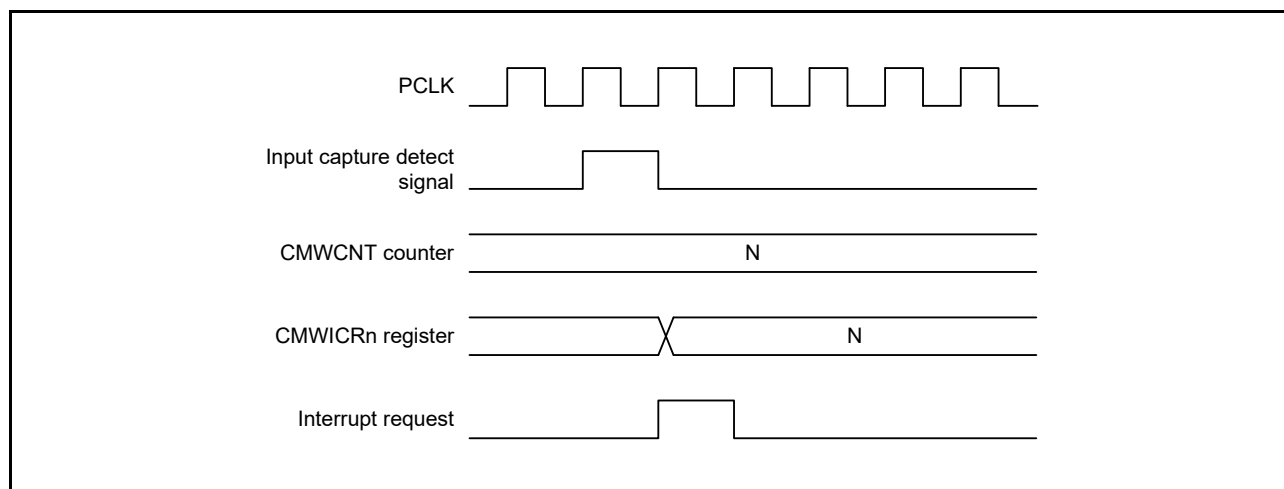


Figure 32.15 Timing of Input Capture Interrupt Generation ($n = 0, 1$)

32.5 Link Operations by ELC

32.5.1 Event Signal Output to ELC

The CMTW uses the event link controller (ELC) to perform link operation to a preset module using the interrupt request signal as the event signal.

The CMTW outputs the event signal upon a compare match. The corresponding channel is channel 0. The event signal can be output regardless of the corresponding interrupt request enable bit (CMWCR.CMWIE).

For details, refer to section 21, Event Link Controller (ELC).

(1) Compare Match Event

In response to a compare match, the CMTW simultaneously issues an interrupt request and a compare match event signal to the ELC. The event signal is issued regardless of the settings of the corresponding interrupt request enable bit (CMWCR.CMWIE bit).

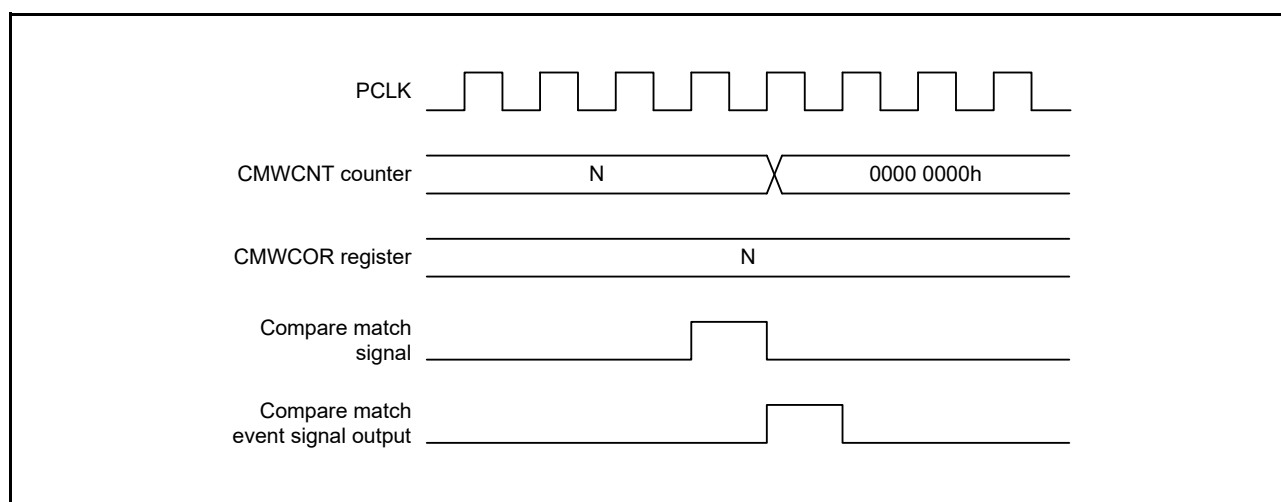


Figure 32.16 Timing of Issuing a Compare Match Event Signal

32.5.2 CMTW Operation When Receiving an Event Signal from ELC

The CMTW can perform any of the following operations according to the event preset by the ELSRn register of the ELC.

(1) Count Start

The CMTW count start operation is selected by the ELOPH register of the ELC. If the event specified by the ELSRn register occurs, the CMWSTR.STR bit is set to 1, starting the CMTW count operation.

However, if the specified event occurs while the CMWSTR.STR bit is 1, the event is ignored.

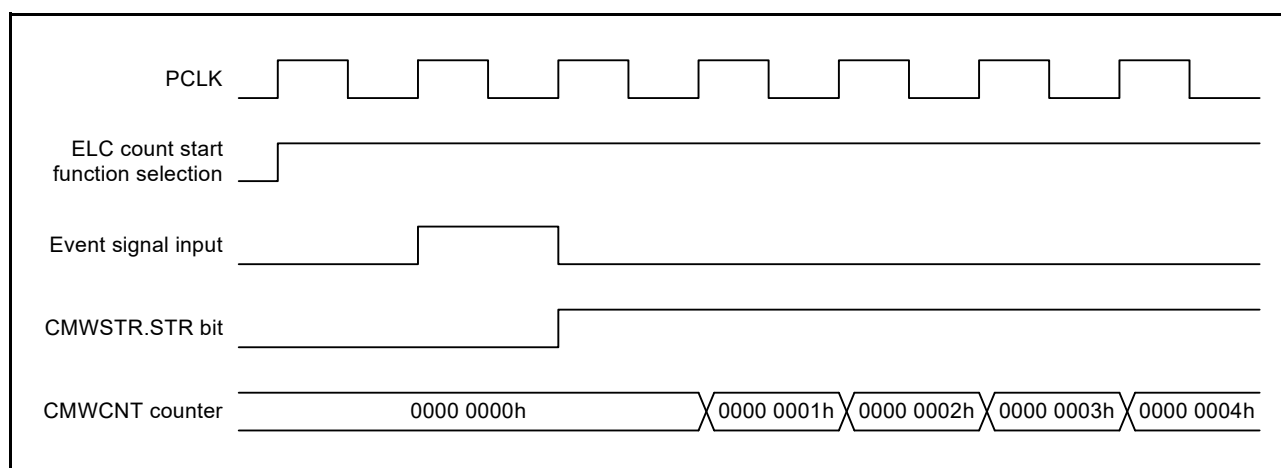


Figure 32.17 Count Start Operation on Acceptance of the Event Signal

(2) Event Count

The CMTW event count operation is selected by the ELOPH register of the ELC. If the event specified by the ELSRn register occurs when the CMWCR.STR bit is 1, the event is counted as the count source regardless of the CMWCR.CKS[1:0] bit setting.

Reading the counter value returns the number of events that have been actually input.

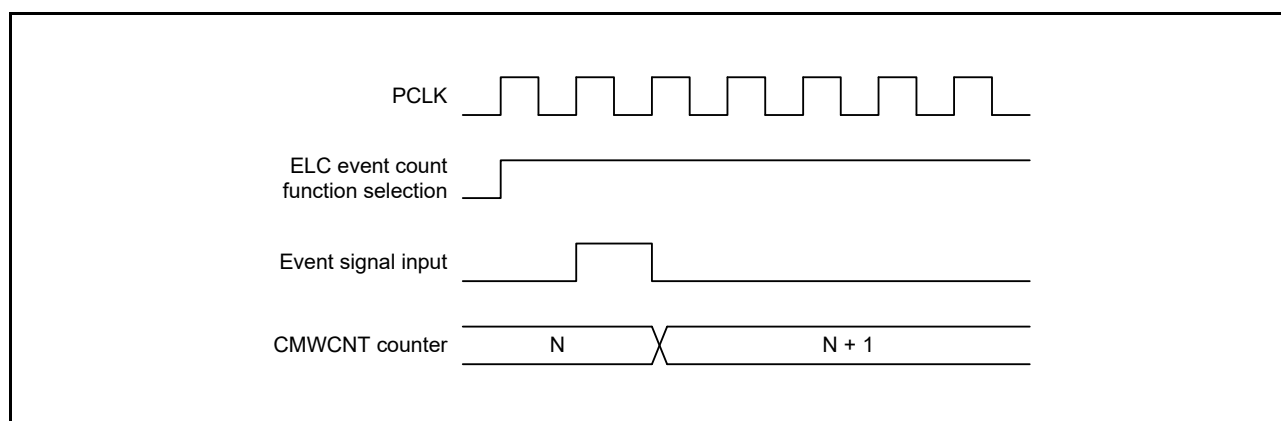


Figure 32.18 Event Count Operation on Acceptance of the Event Signal

(3) Count Restart

The CMTW count restart operation is selected by the ELOPH register of the ELC. If the event specified by the ELSRn register occurs, the CMWCNT counter value becomes 0000 0000h. If the CMWSTR.STR bit is 1, the count operation can be continued.

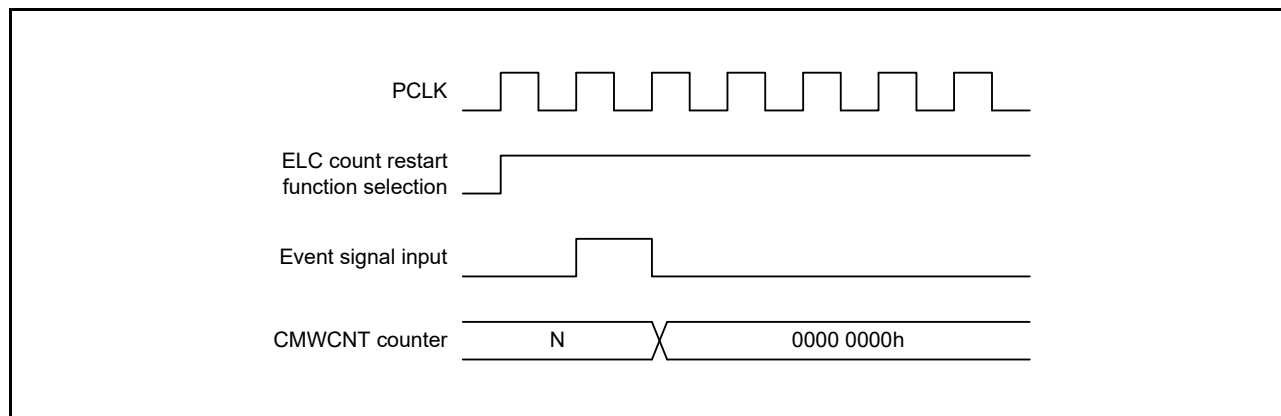


Figure 32.19 Count Restart Operation on Acceptance of the Event Signal

32.5.3 Conflict between Event Link Operation and Register Access

The followings are the notes on using CMTW for event link operations.

Table 32.4 lists count operations when event link operation and register access conflict.

Table 32.4 Count Operations When Event Link Operation and Register Access Conflict (n = 0, 1)

Event Link Operation	Register Access	CMWCNT Counter Status	Operation to be Performed
Count start	Writing to the CMWSTR.STR bit	Stopped state	Count start
		Compare match	Count start
		Counting up	Count start
Event count	Writing to CMWCNT counter	—	Event count
	Writing to CMWCOR register	Compare match	Compare match
Count restart	Writing to CMWCNT counter	Other than compare match	Count restart
	Writing to CMWCNT counter	Compare match	Compare match
	(No access to registers)	Compare match	Compare match
(No events)	Writing to CMWCNT counter	Compare match	Output of compare match interrupt request Writing to CMWCNT counter
		Counting up	Writing to CMWCNT counter
	Writing to CMWCOR register	Compare match	Compare match
	Writing to CMWOCRn register	Compare match	Compare match
	Reading from CMWCNT counter	Counting up	Counting up and reading of the previous value

(1) Count Start

When writing to the STR bit in the CMWSTR register and acceptance of the event signal are in contention, writing to the STR bit does not proceed since setting of the STR bit to 1 in response to the event takes priority.

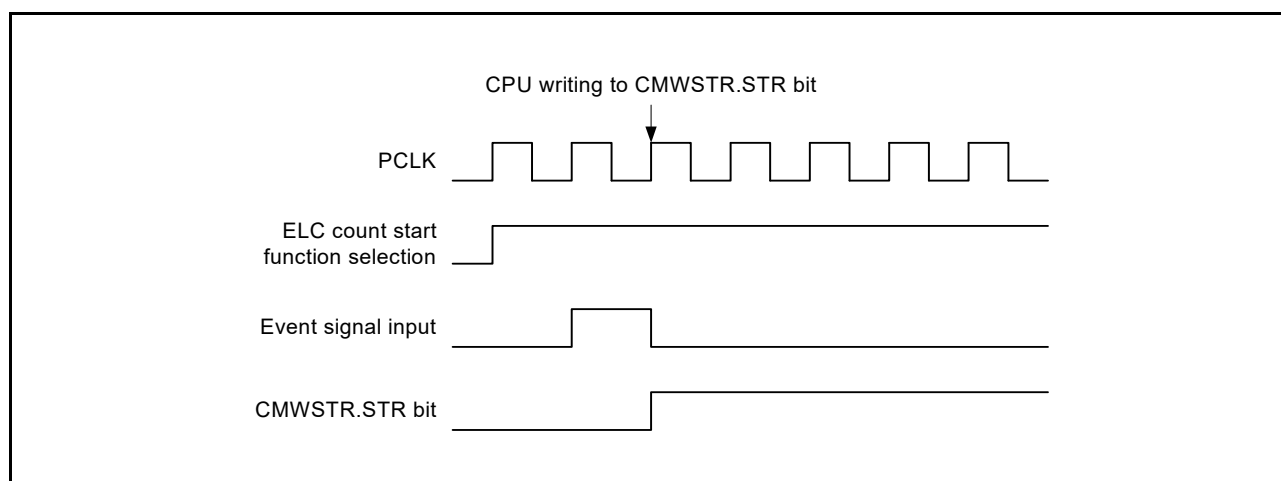


Figure 32.20 Conflict between Event Acceptance and Register Access in Count Start Operation

(2) Event Count

When writing to the CMWCNT counter and acceptance of the event signal are in contention, writing to the CMWCNT counter does not proceed since the count operation in response to the event takes priority.

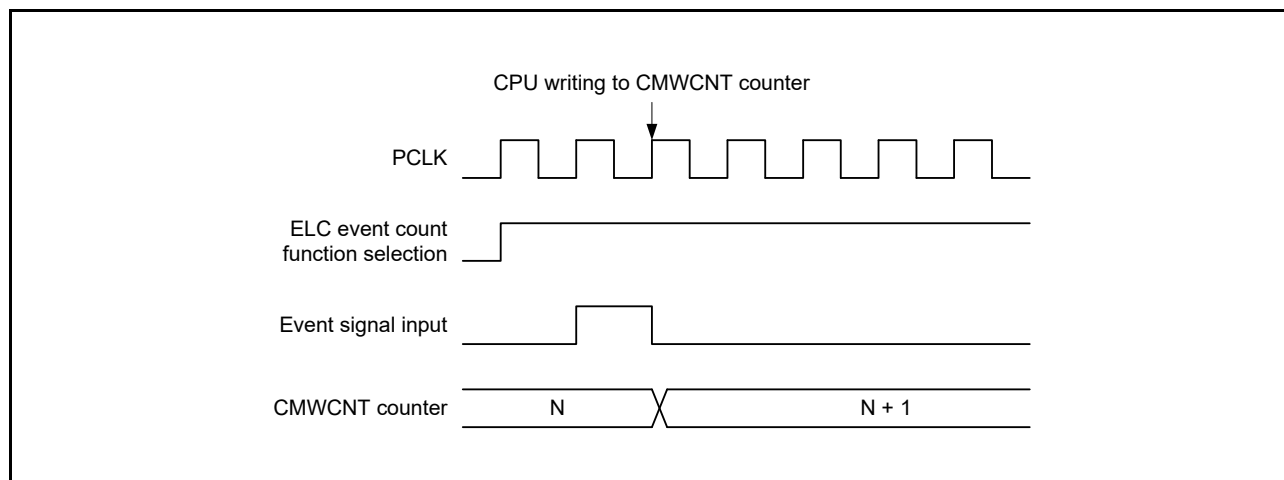


Figure 32.21 Conflict between Event Acceptance and Register Access in Event Counting Operation

(3) Count Restart

When writing to the CMWCNT counter and acceptance of the event signal are in contention, writing to the CMWCNT counter does not proceed since the counter value initialization in response to the event occurrence takes priority.

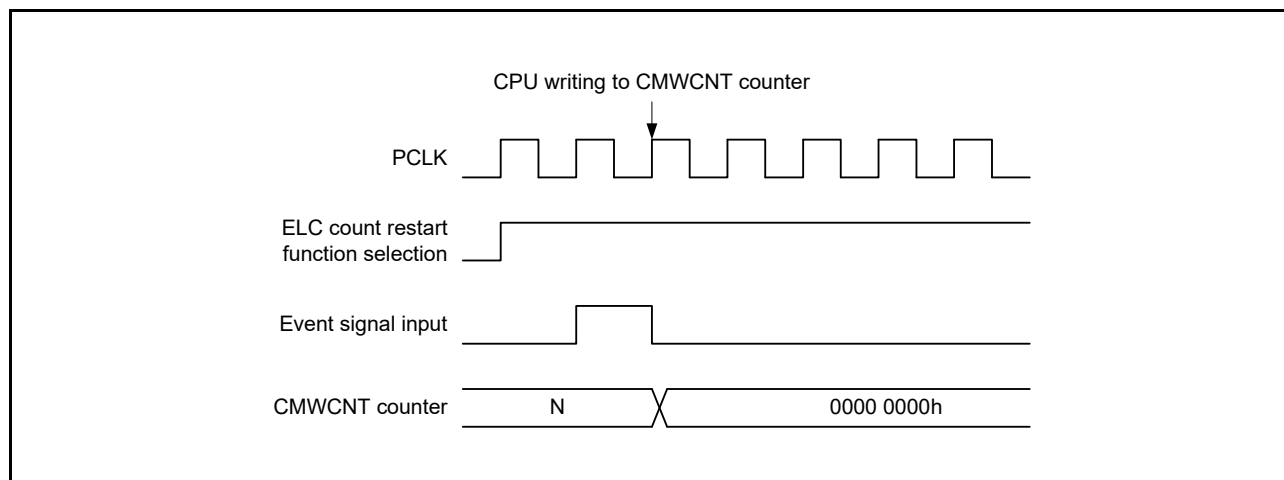


Figure 32.22 Conflict between Event Acceptance and Register Access in Count Restart Operation

32.6 Usage Notes

32.6.1 Setting the Module Stop Function

The CMTW operation can be enabled or disabled using the MSTPCRA register. The CMTW0 and CMTW1 are initially disabled after a reset. Register access is enabled by releasing the module stop state. For details, refer to [section 11, Low Power Consumption](#).

32.6.2 Conflict between CMWCNT Counter Writing and Compare Match

If the compare match signal is generated during writing to the CMWCNT counter, the compare match request is output but the counter is not cleared since writing to the counter takes priority.

Figure 32.23 shows the timing of conflict between CMWCNT counter writing and compare match.

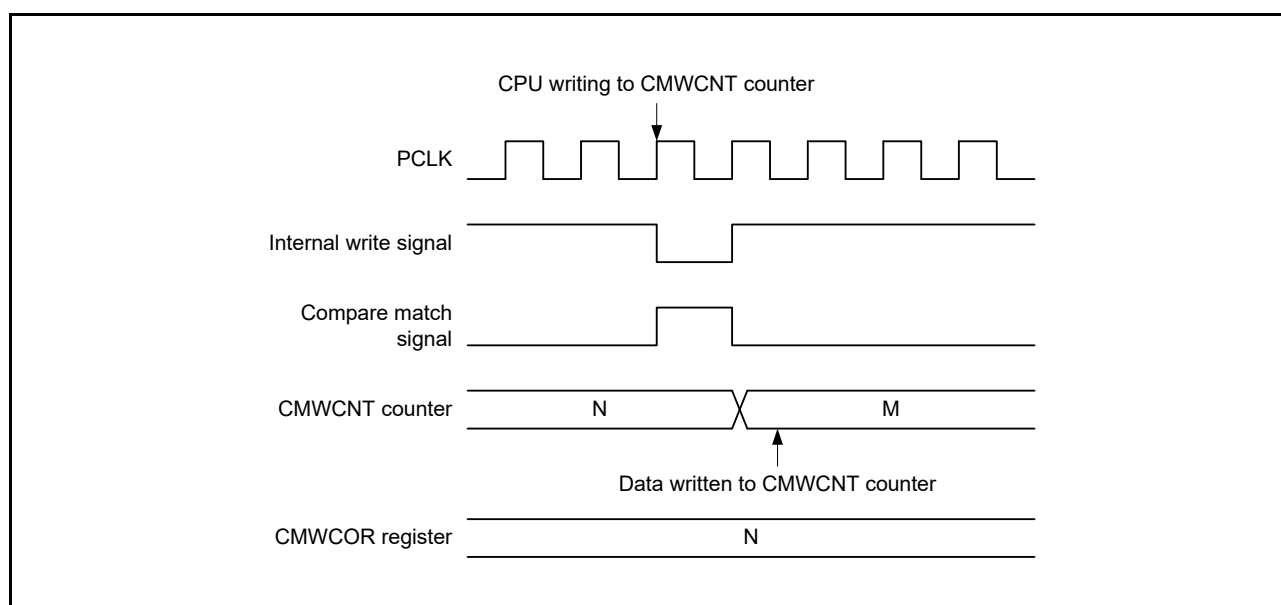


Figure 32.23 Conflict between CMWCNT Counter Writing and Compare Match

32.6.3 Conflict between CMWCNT Counter Writing and Incrementing or Clearing

In case of conflict between incrementation or clearing of the CMWCNT counter and writing to the CMWCNT counter, the counter is not actually incremented or cleared since writing to the CMWCNT counter takes priority.

Figure 32.24 shows the timing in the case of contention between writing to the CMWCNT counter and incrementation or clearing.

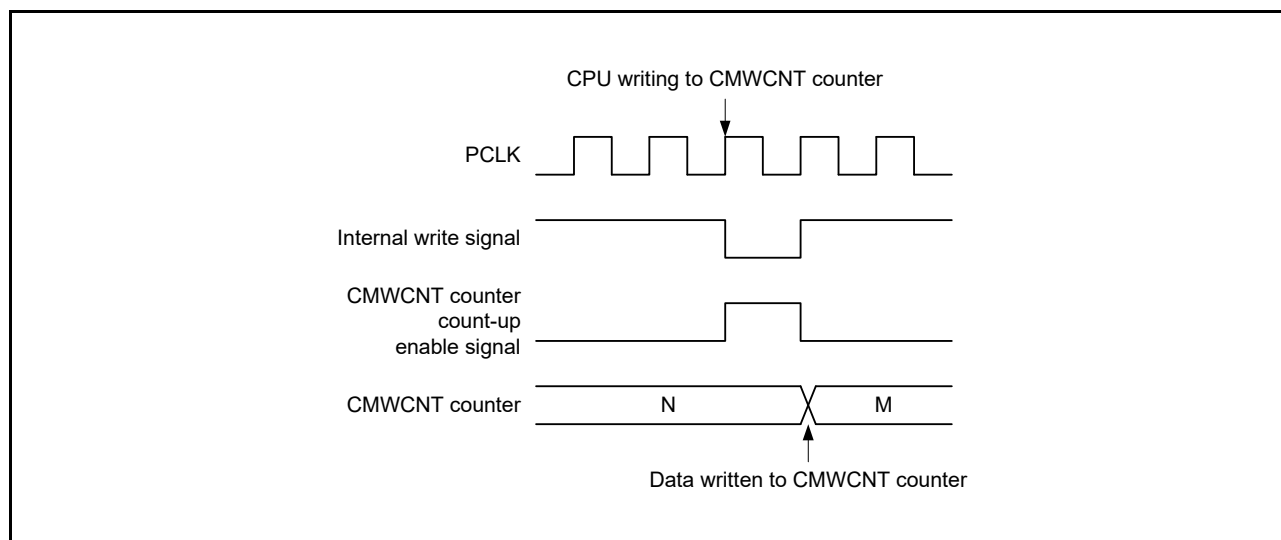


Figure 32.24 Conflict between CMWCNT Counter Writing and Incrementing

32.6.4 Conflict between CMWCOR Register Writing and Compare Match

If the compare match is generated during the CMWCOR register write cycle, the writing to the CMWCOR register takes priority and also the compare match signal is output.

Figure 32.25 shows the timing of conflict between CMWCOR register writing and compare match.

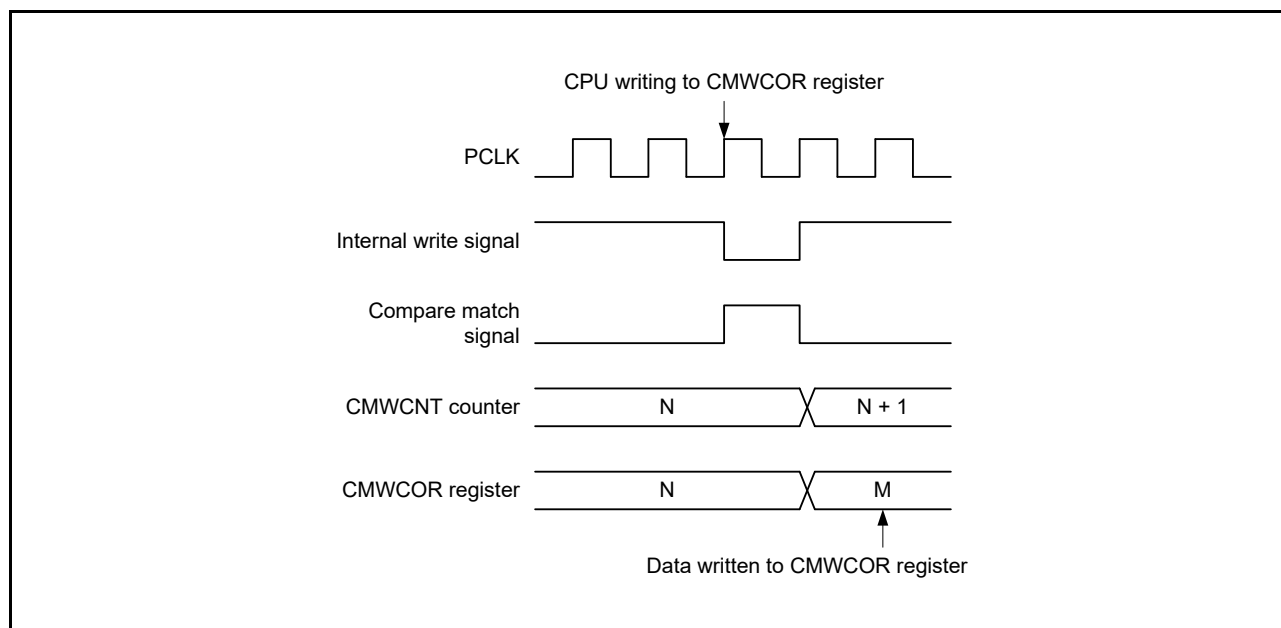


Figure 32.25 Conflict between CMWCOR Register Writing and Compare Match

32.6.5 Conflict between CMWOCRn Register Writing and Compare Match (n = 0, 1)

If the compare match is generated during the CMWOCRn register write cycle, the writing to the CMWOCRn register takes priority and also the compare match signal is output.

Figure 32.26 shows the timing of conflict between CMWOCRn register writing and compare match.

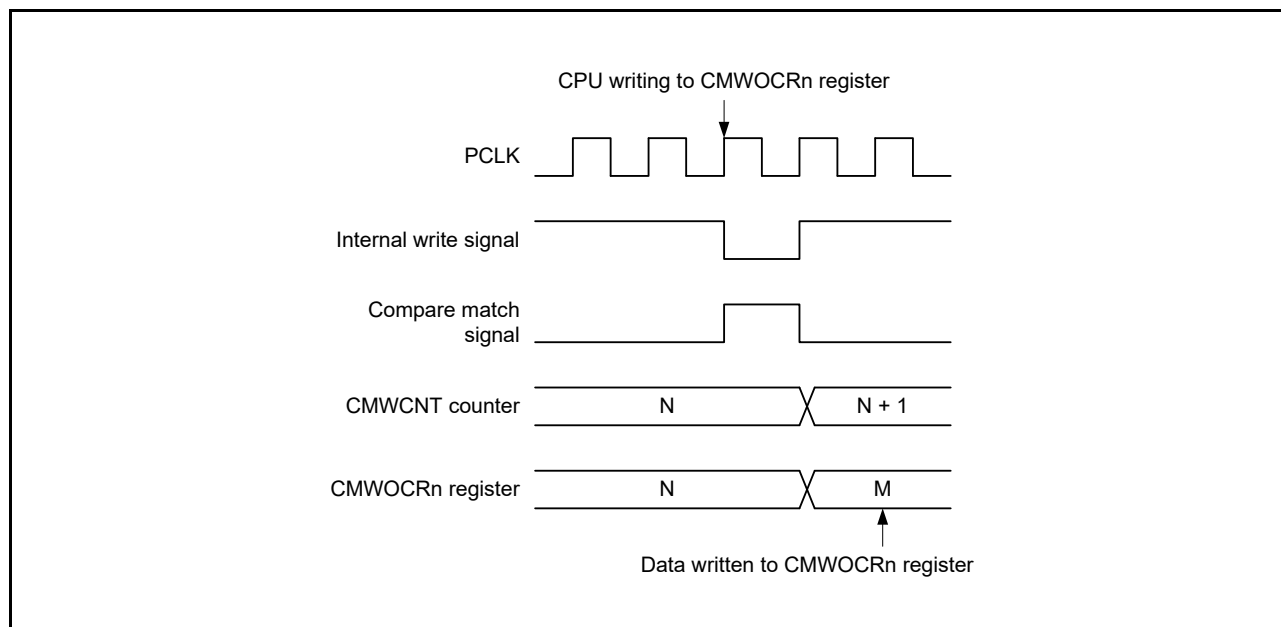


Figure 32.26 Conflict between CMWOCRn Register Writing and Compare Match

32.6.6 Conflict between CMWCNT Counter Reading and Incrementing or Clearing

If the CMWCNT counter incrementing or clearing process occurs at the same time that the data of the CMWCNT counter is read, the value having been in the CMWCNT counter before incremented or cleared is read.

Figure 32.27 shows the timing of conflict between CMWCNT counter reading and incrementing.

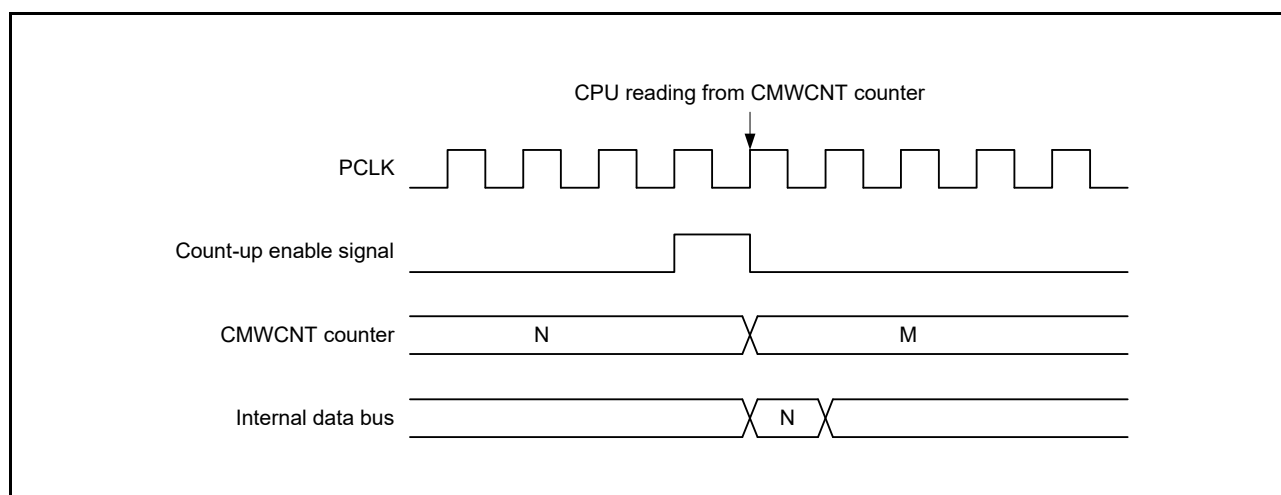


Figure 32.27 Conflict between CMWCNT Counter Reading and Incrementing

32.6.7 Conflict between CMWICRn Register Reading and Input Capture (n = 0, 1)

If the input capture detection signal is generated at the same time that the data of the CMWICRn register is read, the value having been in the CMWICRn register before updated by input capture transfer is read.

Figure 32.28 shows the timing of conflict between CMWICRn register reading and input capture.

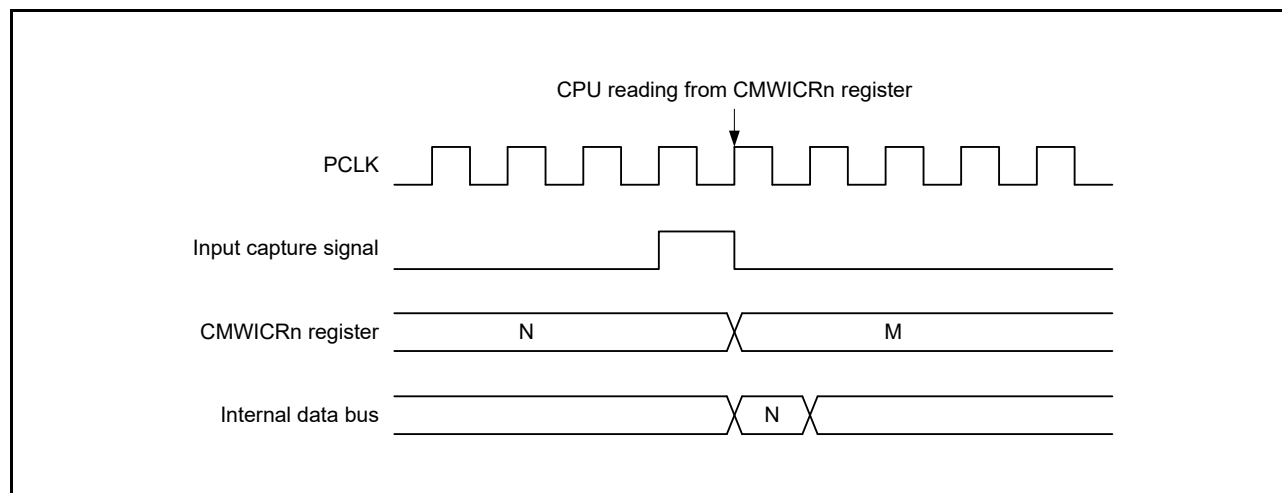


Figure 32.28 Conflict between CMWICRn Register Reading and Input Capture